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PE-PINN: A Physics-Enhanced Physics-Informed Neural Network for Solving the PKN Model of Hydraulic Fracturing*

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This paper presents PE-PINN, a Physics-Enhanced Physics-Informed Neural Network framework for solving the PKN model of hydraulic fracturing. The governing Nordgren equation is solved by embedding physical priors directly into the network architecture, including a transient tip shape function that analytically interpolates asymptotic regimes, a causal arrival-time constraint, and a learnable time-power exponent. Numerical experiments demonstrate competitive accuracy against finite difference solutions while maintaining a mesh-free formulation.

Keywords: Physics-informed neural network; PKN model; hydraulic fracturing; Nordgren equation; moving boundary; tip singularity.

1. Introduction

Hydraulic fracturing is a crucial technology for enhancing hydrocarbon recovery from unconventional reservoirs Economides and Nolte (2000). Accurate prediction of fracture geometry — the spatial and temporal evolution of fracture width, length, and net pressure — is essential for optimizing treatment design and evaluating well performance. Perkins and Kern and Nordgren first proposed the PKN model to describe the evolution of fracture length and width during hydraulic fracturing Nordgren (1972). This classical model, together with the KGD model Geertsma and

*For the title, try not to use more than 3 lines. Typeset the title in 10 pt bold.
†Typeset names in 8 pt Roman.
‡State completely without abbreviations.

de Klerk (1969), established the theoretical basis for the simulation of hydraulic fracturing. Among these models, the PKN model remains one of the most widely adopted in industrial practice due to its computational tractability. The PKN model assumes a fracture of fixed height that is much larger than its vertical extent, with plane-strain deformation in each cross-section. Under this assumption, the local elasticity relation yields a nonlinear governing equation that balances the temporal evolution of fracture width, the fluid leak-off into the surrounding formation, and the longitudinal gradient of the volumetric flow rate governed by the fluid's viscous resistance. Despite its apparent simplicity, this equation poses significant computational challenges due to its strong nonlinearity, moving free boundary, and singular behavior at the propagating tip.

The numerical solution of the PKN model has been an active research topic since Nordgren's original finite-difference formulation Nordgren (1972). Traditional grid-based approaches include explicit finite-difference schemes Kemp (1990), finite element methods with tip enrichment Garikapati *et al.* (2017), and spectral discretizations Peirce (2014). However, these methods face persistent difficulties. The fracture tip exhibits a singular gradient that tends to infinity, requiring special tip elements, regularization techniques Linkov (2011), or asymptotic enrichment functions Garikapati *et al.* (2017) to avoid severe accuracy degradation. The classical PKN tip asymptote follows a one-third power-law in the viscosity-dominated regime Nordgren (1972); Kemp (1990), but a more complex transient behavior — smoothly transitioning from a one-third to a three-eighths power-law depending on the local leak-off intensity — has been recognized in recent re-examinations Linkov (2015); Detournay (2016). Furthermore, the moving free boundary evolves nonlinearly with time, following different power-law regimes depending on whether the propagation is dominated by fluid storage or leak-off. Accurately tracking this evolving crack front typically requires adaptive remeshing or front-tracking algorithms Peirce (2014), which introduce significant computational overhead. Additionally, a multi-scale structure arises from the thin boundary layer near the tip to the global fracture scale, demanding dense local grids that can make parametric studies and real-time applications prohibitive Detournay (2016).

Physics-Informed Neural Networks (PINNs) were introduced by Raissi *et al.* Raissi *et al.* (2019) as a mesh-free framework that embeds the governing partial differential equations directly into the loss function of a neural network. PINNs exploit automatic differentiation for exact derivative computation and can naturally incorporate sparse observational data. Since their introduction, PINNs and their variants — including domain-decomposition extensions such as XPINN Jagtap and Karniadakis (2020) and cPINN Jagtap *et al.* (2020), gradient-enhanced formulations Yu *et al.* (2022), and multi-scale architectures Moseley *et al.* (2023) — have been successfully applied across a broad spectrum of problems in computational mechanics. Recently, PINNs have been applied to the field of subsurface engineering, including flow and transport in porous media Hassan and Labak (2024), history-matching in fractured reservoirs Ali *et al.* (2025), and two-phase flow in heterogeneous formations Wang *et al.* (2024). However, the application of PINNs to hydraulic fracture propagation — characterized by its unique combination of a moving free boundary, singular tip asymptotics, and multi-regime transient behavior — remains relatively unexplored.

A small number of recent studies have begun to bridge PINNs with hydraulic fracture modeling. Yang *et al.* (2023) developed a physics-informed surrogate model for predicting hydraulic fracture geometry using deep learning, demonstrating the feasibility of replacing computationally expensive simulators with trained neural networks. Tang *et al.* (2024) proposed a PINN formulation with moving boundary constraints for modeling hydraulic fracturing, making an important step toward handling the free-boundary nature of the problem. However, existing PINN-based approaches for hydraulic fracturing still face critical limitations. First, most existing PINN formulations for fracture problems do not explicitly encode the asymptotic tip behavior into the network architecture. Without such prior knowledge, the network must learn sharp near-tip gradients purely from PDE residuals, which is notoriously difficult for gradient-based optimization Krishnapriyan *et al.* (2021). The transient transition between the one-third and three-eighths power-law tip asymptotics has not been addressed in any existing PINN work. Second, the PKN governing equation is physically valid only in the region where the fracture has already arrived. Existing PINN formulations do not enforce this causal constraint, leading to non-physical residuals being minimized in regions not yet reached by the fracture front. Third, the early-time behavior of fracture width exhibits a power-law singularity whose exponent depends on the dominant physical regime. Learning this temporal scaling directly from data is inefficient and limits generalization to time ranges not covered during training.

To address these challenges, this paper proposes PE-PINN (Physics-Enhanced PINN), a novel framework for the PKN model that explicitly embeds physical prior knowledge into the network architecture. The proposed method decomposes the fracture width into three physically meaningful components: a learnable time-power term that captures the early-time singular scaling, an analytically derived transient tip shape function that smoothly interpolates between the one-third and three-eighths asymptotic regimes, and a neural network that learns the remaining smooth component. This physics-enhanced representation explicitly separates the singular time and space dependencies, allowing the network to learn only a well-behaved residual function. Building on this decomposition, the arrival time of the crack tip at each spatial location is computed by numerically inverting the crack-length evolution relation, and the PDE residual evaluation is restricted to the physically meaningful region where the fracture has already arrived. This causal masking ensures physical consistency. Additionally, the time-power exponent is constrained to a physically plausible range and optimized jointly with the network parameters, automatically discovering the correct temporal scaling. A two-stage training strategy combining Adam Kingma and Ba (2015) for global exploration with L-BFGS Liu and Nocedal (1989) for fine local convergence is employed. In contrast to existing PINN formulations for hydraulic fracturing Yang *et al.* (2023); Tang *et al.* (2024), the proposed method encodes the asymptotic tip structure analytically rather than relying on the optimizer to discover it, which significantly reduces the learning burden on the network. Numerical experiments demonstrate that the proposed method achieves competitive accuracy against conventional finite difference solutions while maintaining a mesh-free formulation. Systematic ablation studies and sensitivity analyses further confirm that each component of the proposed framework contributes measurably to overall accuracy.

The remainder of this paper is organized as follows. Section 2 formulates the PKN governing equations. Section 3 describes the proposed PE-PINN framework, including the network design, loss function, training strategy, and experimental configuration. Section 4 presents the numerical results. Section 5 concludes the paper.

2. Problem Formulation

The PKN model describes the propagation of a hydraulic fracture of constant height H within a homogeneous, linear elastic reservoir Nordgren (1972). The fracture is assumed to be much longer than its height, resulting in plane-strain deformation in each vertical cross-section perpendicular to the propagation direction. The fluid flow within the fracture is laminar and governed by lubrication theory. The fracture propagates in the viscosity-dominated regime, and fluid leak-off from the fracture faces into the surrounding formation follows Carter's one-dimensional leak-off model Detournay (2016).

Under the plane-strain assumption, the net pressure $p(x,t)$ is proportional to the local fracture width $W(x,t)$ at each vertical cross-section Nordgren (1972):

$$p(x,t) = \frac{E'}{4H}\,W(x,t), \tag{1}$$

where $E' = E/(1-\nu^2)$ is the plane-strain Young's modulus. The fluid flow within the fracture is described by the lubrication equation, which relates the local flow rate to the pressure gradient. Combining the elasticity and lubrication equations with the mass conservation condition that accounts for Carter's leak-off yields the Nordgren equation:

$$\frac{\partial W}{\partial t} + \frac{C_L}{\sqrt{t-\tau(x)}} - \frac{E'}{256H^2\mu}\,\frac{\partial^2(W^4)}{\partial x^2} = 0, \tag{2}$$

where C_L is the leak-off coefficient, μ is the fluid viscosity, and $\tau(x)$ is the arrival time of the fracture tip at the spatial coordinate x. The boundary condition at the wellbore ($x = 0$) relates the injection rate to the fracture width gradient through the flux condition. At the propagating tip ($x = L(t)$), the fracture width vanishes.

The fracture length $L(t)$ evolves nonlinearly with time, following distinct power-law regimes in the storage-dominated ($L \propto t^{4/5}$) and leak-off-dominated ($L \propto t^{1/2}$) limits Nordgren (1972). A harmonic mean interpolation smoothly connects these asymptotic limits.

A critical feature of the PKN model is the singular behavior of the fracture width near the propagating tip. In the classical viscosity-dominated regime, the width follows a one-third power-law $W \sim (1-x/L)^{1/3}$ as $x \to L(t)$ Nordgren (1972); Kemp (1990). However, a more general analysis reveals a transient tip shape that smoothly transitions from the one-third power-law to a three-eighths power-law depending on the local balance between fluid storage and leak-off Linkov (2015). This transient behavior is characterized by a dimensionless tip shape function $\Omega_{\text{tip}}(\xi)$, where $\xi = 1 - x/L(t)$ denotes the dimensionless distance from the tip. Accurately capturing this transition is essential for resolving the near-tip solution without excessive mesh refinement.

3. Methodology

3.1. *PE-PINN design*

Physics-Informed Neural Networks (PINNs) were introduced by Raissi *et al.* Raissi *et al.* (2019) as a mesh-free framework for solving partial differential equations. A fully connected neural network approximates the solution $u(x, t; \theta)$, and the governing PDE residual is embedded directly into the loss function. Automatic differentiation is employed to compute the required derivatives exactly.

The proposed PE-PINN framework addresses three fundamental challenges that conventional PINNs face when applied to the PKN model with a moving boundary and singular tip behavior.

Physics-enhanced ansatz. Rather than directly approximating the fracture width with a generic neural network, the proposed method decomposes the solution into three physically meaningful components. The time-power factor captures the early-time singular scaling, the transient tip shape function analytically encodes the asymptotic crack-tip behavior, and the neural network learns the remaining smooth component. This decomposition explicitly separates the singular dependencies, allowing the network to learn only a well-behaved residual function.

Causal arrival-time constraint. The PKN governing equation is physically valid only in the region where the fracture has already arrived. The proposed method computes the arrival time $\tau(x)$ at each spatial location by numerically inverting the $L(t)$ relation using Brent's root-finding method. The PDE residual is evaluated only where $t > \tau(x)$ and $x \leq L(t)$, ensuring that the network is not penalized for violating physics in regions not yet reached by the fracture front.

Learnable time-power exponent. The early-time temporal scaling exponent is treated as a trainable parameter constrained to a physically plausible range. This parameter is optimized jointly with the network weights, automatically discovering the correct scaling without manual tuning. Its converged value provides

Table 1. Physical parameters.

Parameter	Symbol	Value
Plane-strain Young's modulus	E'	[TBD]
Fracture height	H	[TBD]
Fluid viscosity	μ	[TBD]
Leak-off coefficient	C_L	[TBD]
Injection rate	Q_0	[TBD]
Simulation time	$T_{\max}$	[TBD]

interpretable physical insight into the dominant flow regime.

Network architecture. The neural network component consists of a fully connected feed-forward architecture with four hidden layers of 128 neurons and Tanh activation. The output is constrained to be positive through a Softplus activation.

3.2. *Loss function and training*

The total loss function comprises the PDE residual loss and the boundary condition loss. The PDE residual is computed only within the causally valid region. A smooth temporal factor is introduced at the wellbore to reconcile the initial and boundary conditions at the origin.

Collocation points are generated using Latin Hypercube Sampling, with additional refinement in the near-tip region to resolve the sharp gradients. Training proceeds in two stages: the Adam optimizer with exponential learning rate decay for global exploration, followed by the L-BFGS optimizer with strong Wolfe line search for fine convergence.

3.3. *Experimental configuration*

The physical parameters adopted in the numerical experiments are summarized in Table 1.

A finite difference (FD) solver is implemented as the reference baseline. The FD scheme employs a uniform spatial grid with explicit time stepping and a velocity-based tip propagation condition.

To quantify the contribution of each component, four model variants are constructed for ablation analysis: the full PE-PINN, PINN, a variant without the time-power term, a variant without the transient tip shape function, and a plain PINN without any of the proposed components. In addition, a systematic sensitivity analysis is conducted by varying the number of hidden layers (depth $D = 2, 3, 4, 5, 6$) and the number of neurons per layer (width $W = 32, 64, 128, 256, 512$), yielding 25 configurations.

4. Results and Discussion

4.1. *Model validation*

The proposed method is evaluated against the finite difference baseline. Fracture width profiles at representative times and the crack length evolution are compared.

4.2. *Ablation study*

The ablation study quantifies the individual contribution of each component of the proposed framework.

4.3. *Sensitivity analysis*

The sensitivity analysis characterizes the influence of network depth and width on prediction accuracy.

5. Conclusion

This paper presented PE-PINN, a Physics-Enhanced PINN framework for the PKN model of hydraulic fracture propagation. The proposed method addresses three fundamental challenges of applying PINNs to moving boundary problems: the transient tip shape function analytically encodes the transition between asymptotic regimes, the causal arrival-time constraint restricts the PDE residual to the physically valid region, and the learnable time-power exponent adaptively captures the temporal scaling behavior. Numerical experiments demonstrate competitive accuracy against a conventional finite difference solver while maintaining a mesh-free formulation. Future work will extend the framework to more complex fracture propagation scenarios.

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